

Muhammad Zaid Mohsin

+92 3259518400 | zaidmohsin45@gmail.com | [LinkedIn: Muhammad Zaid Mohsin](#) | [GitHub: ZaidMohsin457](#)

EDUCATION

Ghulam Ishaq Khan Institute - Bachelor of Science in Computer Science

September 2022 – October 2026

CGPA: 3.66 GIK Merit Scholarship Winner, **3-time** Dean's Honar Award Winner

Relevant Coursework

Introduction to Artificial Intelligence, Cloud Computing, Database Management System, Computer Communication and Networks, Operating Systems, Data Structures and Algorithms, Software Engineering.

PROFESSIONAL AND LEADERSHIP EXPERIENCE

Backend Developer, Talpur Foundation Pakistan

January 2025 – February 2025

- Designed and developed scalable **Django-based backend architecture**, implementing RESTful APIs and database optimizations for enhanced performance.
- Developed and optimized Django-based APIs, improving backend **response time by 40%** and reducing **database queries by 30%**.
- Orchestrated the deployment and hosting on **Vercel**, delivering a highly reliable and scalable platform.
- Launched the live website at <https://talpurfoundationpakistan.com/>, exemplifying modern web development practices.

Lead Organizer, AWS Pakistan Session– GIKI

September 2024– Present

- Led a collaboration between AWS Pakistan and GIKI, organizing a workshop on **AI in web development and AWS services**, strengthening industry-academia ties and established relations for future collaboration with AWS Pakistan for the first time.
- Coordinated event logistics and managed cross-functional teams for a workshop attended by **40+ participants**, showcasing strong organizational and leadership abilities relevant to supply chain management.
- Successfully managed** event logistics, including venue coordination, participant registration, and session execution, contributing to a **highly rated and well-received event**.

Machine Learning Internee, Digital Empowerment Network

July 2024 – August 2024

- Completed a 4-week internship at Digital Empowerment Network (DEP) as a Machine Learning Intern, gaining hands-on experience in practical machine learning applications
- Utilized machine learning for data analysis and process optimization, skills that translate to effective supply chain analytics.

Society of Mechanical Engineering Pakistan, GIKI

January 2023 – Present

- Organized **3 HEC-certified workshops** in Swabi universities, attended by **150+ students**.

PROJECTS

AI Powered Customer Relationship Model (CRM) Database Management System

Python, Streamlit, OpenAI, PostgreSQL

- Developed a CRM system that reduced **administrative overhead by 40%**, reflecting an ability to streamline processes and enhance operational efficiency—key competencies in supply chain management
- Implemented an NLP-driven chatbot within the CMS, capable of **resolving 75% of user queries** related to project updates, employee details, and client information in real time, **decreasing manual support efforts by 50%** and enhancing user productivity.
- Streamlined backend workflows by **reducing response time by 40% and cutting database queries by 30%**, demonstrating skills in process optimization similar to supply chain streamlining.

AI-Based Course Recommendation System Artificial Intelligence

Python, Streamlit, OpenAI

- Proceeded with a personalized course recommendation system utilizing Random Forest and content-based filtering, achieving an **RMSE of 0.258 and MSE of 0.066** in predicting user ratings for Coursera courses, delivering highly accurate and tailored learning suggestions.
- Engineered a recommendation system using **data analytics to personalize user experience**, demonstrating proficiency in forecasting and resource allocation applicable **to supply chain demand planning**.
- Enhanced the online learning experience by providing precise course recommendations based on user preferences and educational goals, potentially cutting course selection time by up to **50%, improving efficiency** and relevance for learners.

Deepfake Video Detection Tool Using Neural Networks Computer Vision

Python, Django, TensorFlow, OpenCV

- Initiated a deepfake detection system using **ResNext CNN for feature extraction and LSTM RNN** for sequence classification.
- Trained model on **6000+ videos (FaceForensics++, Celeb-DF)**, achieving **97.76% accuracy on FaceForensics++ with 100-frame sequences**.
- Architected a scalable video processing model that efficiently analyzes sequences ranging from 10 to 100 frames, **delivering 89.35% accuracy** on a mixed dataset with 40 frames, optimized for real-time performance across various video types and resolutions.

RESEARCH EXPERIENCE

Enhancing IoT Security with Machine Learning

- Developed a scalable IoT security framework with federated and adaptive learning techniques, demonstrating the ability to design systems that optimize and monitor complex networks, similar to supply chain operations.
- Integrated federated and adaptive learning to boost anomaly detection and reduce bandwidth consumption.

- Conducted comprehensive literature reviews and experimental validations to address real-world security challenges.

Innovative Facial Emotion Detection Using Linear Algebra & Control Charts

- Applied Singular Value Decomposition (SVD) for efficient dimensionality reduction in high-dimensional facial data.
- Leveraged statistical process control (Shewhart, EWMA, and CUSUM charts) to monitor system stability and detect anomalies.
- Enhanced feature extraction methods to improve the accuracy and sensitivity of emotion recognition systems.
- Applied statistical process control techniques to monitor system stability and detect anomalies, mirroring quality control and continuous improvement processes in supply chain management.

Comprehensive Analysis of Deadlock Handling Methods in Operating Systems

- Analyzed critical conditions for deadlock occurrences and proposed innovative approaches to enhance system reliability.
- Provided actionable recommendations for improving concurrency and performance in multi-process environments.
- Evaluated deadlock handling strategies to optimize resource allocation and enhance system reliability, paralleling the challenges of managing resource flows in supply chains.

TECHNICAL SKILLS

Languages: C++, C, HTML, Python, SQL, Verilog, Assembly Language, Bash.

Tools and Technologies: AWS, Azure, Vercel, Linux (Ubuntu), Windows, Git, Jupyter Notebook, PostgreSQL, LaTeX/Overleaf,

Soft Skills & Leadership: Public Speaking, Teamwork, Leadership, Content Writing.

Libraries: pandas, NumPy, Matplotlib, tensorflow, OpenCV.

Frameworks & Tools: Django, Flask

AI & Machine Learning: NLP, APIs, LLMs, Supervised Learning, Unsupervised Learning, Model Deployment,